Supplementary Material:

Every Fact, and how it was obtained

for *Inflation With and Without an Internal Space: Exact Arithmetic of the Spectre Monotile and the Chair44*

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What this is

The main paper stratifies its statements by epistemic weight: *Facts* are computations or theorems, *Propositions* carry one modelling step, *Conjectures* are sharp and unproved. This document lists every Fact — all 90 — with the key under which `spectrefacts.py` computes it, the claim in the form the paper states it, the first sentence of the method by which it was obtained, and a mark on the 27 Facts that are also proved by the Lean 4 kernel.

Three things about the table.

- **It is generated, not written.** Every row is emitted at build time from `spectrefacts.py`, the same module that produces every number in the main paper. There is no second copy to drift.
- **The Lean column is measured, not asserted.** A checkmark means the Fact's `lean` field names a theorem that exists in `SpectreFacts.lean` — 67 theorems, kernel-accepted, no admitted proofs — and the paper's build fails if a named theorem is missing. Six such phantom attributions were found and removed when that check was added; the count of checked Facts fell from twenty-two to eighteen, and eighteen is the honest number.
- **The method column is the first sentence.** A method that runs to a paragraph is truncated at its first full stop or semicolon. The full text is in the module, one lookup by key away.

Two of the Facts are inputs rather than results, and the paper says so: `charpoly` rests on the transcribed substitution matrix, and `boundary_recurrence` on a perimeter seed counted from placed tiles. Both are audited by an independent route — the matrix against a second implementation of the same substitution, the seed against an exact $\mathbb{Q}(\sqrt{3})$ recount — and an audit is not a proof. They are the two rows a sceptical reader should look at first.

The table

Generated from `spectrefacts.py`. The last column marks the Facts that are also proved in Lean, see https://github.com/brentharts/spectre/wiki/Spectre-Lean; the others are computations.

Key	Claim	How	Lean
`charpoly`	$\chi_M(x) = x^5(x-1)(x+1)(x^2-8x+1)$	exact characteristic polynomial over Z, factored	
`perron`	$\lambda^2 = 4+\sqrt{15}$	largest root of the quadratic factor	
`pell`	$4^2 - 15 \cdot 1^2 = 1$	the Pell equation, so lambdaˆ2 is the fundamental unit of Z[sqrt15]	✓
`lam_closed`	$\lambda = \frac{1}{2}(\sqrt{6}+\sqrt{10})$	square both sides: (sqrt6+sqrt10)ˆ2/4 = (16+4 sqrt15)/4	
`minpoly`	$x^4 - 8x^2 + 1$	minimal polynomial of lambda over Q	
`channel_hex`	$(\lambda - \lambda^{-1})^2 = 6$	lambdaˆ2 + lambdaˆ-2 - 2, in Z[sqrt15], no root evaluated	✓
`channel_pent`	$(\lambda + \lambda^{-1})^2 = 10$	lambdaˆ2 + lambdaˆ-2 + 2, in Z[sqrt15], no root evaluated	✓
`contraction`	$\lambda^{-4} = 31 - 8\sqrt{15}$	the square of the conjugate unit	✓
`jordan`	$\operatorname{rank} M = 5$, $\operatorname{rank} M^n = 4$ $(n \geq 2)$	exact elimination over Q	

Key	Claim	How	Lean
`q_plus_eigen`	$Q_+ M = Q_+$	left eigenvector for the marginal eigenvalue +1	✓
`q_minus_eigen`	$Q_- M = -Q_-$	left eigenvector for the marginal eigenvalue -1	✓
`q_values`	$Q_+ \equiv -1, \quad Q_- = +1, -1, +1, -1$	evaluated on the matrix census at depths one to four	✓
`censuses`	$T = 1, 8, 63, 496$; $T_{\text{geom}} = 1, 9, 71, 559$	the same patches counted with and without the Mystic	
`mystic_rule`	$T_{\text{geom}}(n) = T(n) + T(n-1)$	row Gamma of M is all ones, so N_Gamma(n) is the whole census at n-1 and the Mystic is a second copy of it	✓
`ledger`	$N_\Theta(n) = N_\Gamma(n-1)$	row Theta of M is the indicator of Gamma, so the count is the previous generation of Gamma verbatim	
`ledger_lambda`	$N_\Lambda(n) = N_\Sigma(n-1)$	row Lambda of M is the indicator of Sigma	
`hat_charpoly`	$\chi_H(x) = (x-1)(x+1)(x^2 - 7x + 1)$	characteristic polynomial of the four-metatile Hat substitution	
`hat_perron`	$\kappa = \frac{1}{2}(7 + 3\sqrt{5}) = \varphi^4$	largest root of the Hat quadratic, and the fourth power of the golden ratio	
`phase_fields`	$\mathbb{Q}(\sqrt{15})$ against $\mathbb{Q}(\sqrt{5})$	the two characteristic polynomials differ in a single coefficient, seven against eight, and that...	
`frequencies`	$v \in \mathbb{Z}[\sqrt{15}]^9$, $\sum v = 1$	nullspace of M - lambdaˆ2 I, normalised	✓
`hat_frequencies`	$f_H = \frac{1}{3}$; $f_T, f_P, f_F \notin \mathbb{Q}$	Perron eigenvector of the four-metatile Hat substitution, over Q(sqrt5)	
`aperiodicity_witness`	9/9 vs 3/4 irrational	a periodic tiling has rational frequencies, so an irrational one refutes periodicity	✓
`trace_accounting`	$\text{tr} = \text{unit} + \text{conjugate} :\ 8\ ,\ 7$	the marginal pair +1 and -1 cancels and the zero modes contribute nothing, so the trace is the unit plus...	✓
`minors`	$D_m = (-1)^m (m+1)$	leading principal minors of the tridiagonal (-2,1,1) form, in closed form	✓
`signature`	$\sigma(T(2,k)) = -(k-1)$, $u(T(2,k)) = \frac{k-1}{2}$	the minors alternate in sign, so the form is negative definite by Sylvester	
`area_form`	$A(a,b) = 2\sqrt{3}\,a^2 + 3ab + \sqrt{3}\,b^2$	symbolic shoelace on the published vertex construction, an identity in Q(sqrt3)[a,b] rather than a fit to...	
`area_endpoints`	$A(1,\sqrt{3}) = 8\sqrt{3}$, $A(\sqrt{3},1) = 10\sqrt{3}$	the Hat and Turtle endpoints of the Tile(a,b) family	
`mystic_asymmetry`	$A(b,a) - A(a,b) = \sqrt{3}\,(b^2 - a^2)$	the exact content of ”only Gamma moves”: the area form is not symmetric, and the Mystic is the species...	
`order_parameter`	$\max \lvert f_\Gamma - f_\Delta \rvert = \frac{35}{67} - \frac{20}{201}\sqrt{15}$	the splitting maximum at the Hat endpoint	
`rho_x`	$\rho_X(p)$ with $p = \Pr(\Phi^2)$ the only unknown	bulk X-density from the per-species X-charges	
`x_charge`	$X_{\Gamma_2} = 14$; $X_{\Gamma_1} = X_\Delta = X_\Sigma = 4$; $X_\Lambda = 2$; $X_\Phi \in \{0, 2\}$	built the tiling and counted, for every interior tile, the shared edges whose two canonical slot roles...	✓
`atlas`	131 contact classes, stable	the unordered adjacency classes (species and slot on each side) number 131 at depths three, four, five and...	
`phi_drift`	$p = \Pr(\Phi^2) :\ 0.683,\ 0.579,\ 0.538,\ 0.519$	the flavour split by depth three to six	
`rho_x_at_half`	$p = \frac{1}{2} \Longrightarrow \rho_X = \frac{2g}{1+g} = 1 - \frac{\sqrt{15}}{5}$	conditional, not established: repeated Aitken extrapolation of the depth three to six values gives 0.5107...	
`chiral_angle`	$\cos\theta_{\text{chiral}} = \frac{1}{8}(1 + 3\sqrt{5})$	Moritake et al	
`angle_from_kappa`	$\cos\theta_{\text{chiral}} = \frac{1}{4}(\kappa - 3)$	3 phi - 1 and phiˆ4 - 3 are the same number, so the measured chiral angle is a function of the Hat...	✓
`spectre_angle_differs`	$(\lambda^2 - 3)/4 > 1$	substituting the Spectre inflation factor into the Hat formula gives a cosine greater than one, so the...	
`twist_two_levels`	$\chi_B(x) = (x-1)^2(x^2 - 8x + 1)$, $B\,v = \lambda^2 v$	the supertile quadrilateral moves by an exact map, transcribed from the construction	
`spectre_twist`	$\tan\theta_S = \frac{1}{13}(5\sqrt{3} - 6)$, $\cos 2\theta_S = \frac{1}{10}(4 + 3\sqrt{3})$	one level is a similarity of ratio lambda with rotation minus theta_S on even levels and plus theta_S on...	
`twist_field`	$\cos 2\theta_S \in \mathbb{Q}(\sqrt{3})$, $\notin \mathbb{Q}(\sqrt{15})$	the angle lies in the field of the hexagonal tile coordinates	

Key	Claim	How	Lean
`twist_family`	$\theta_S(b/a)$ varies continuously with the tile shape	the same derivation for Tile(1, r): the rotation depends on the quadrilateral and so on the shape, passing...	
`bh_theorem`	$u(7_1\#\overline{7_1}) \leq 5 < 6 = u(7_1) + u(\overline{7_1})$	Brittenham and Hermiller 2025, arXiv:2506.24088, Theorem 1.2: the first failure of additivity of...	
`bh_version`	cite v2, not v1	the first version routed through two diagrams of K15n81556 asserted to be the same knot	
`bh_gap_was_chirality`	the gap was a mirror confusion	the two diagrams differed by mirroring, which is exactly the distinction the binding mechanism turns on	
`bh_deficit_bound`	$1 \leq \delta \leq 4$	the deficit is bounded, not determined	
`bh_threshold`	$k, \ell \geq 7$ odd	their Corollary 1.3 covers T(2,2k+1) with k ¿= 3, that is index seven and above	
`beta`	$\beta_{\text{pred}} = \frac{70}{67} - \frac{40}{201}\sqrt{15}$	the order-parameter maximum read in degrees under the stated unit postulate – a modelling step, not a...	
`beta_tension`	$\lvert\beta_{\text{pred}} - \beta_{\text{obs}}\rvert/\sigma$	against the 2026 joint ACT DR6 + Planck PR4 value 0.277 +- 0.057 degrees	
`boundary_recurrence`	$P(k) = 5P(k-1) - 3P(k-2) - P(k-3)$	exact integer recurrence on the supertile perimeter, derived in spectre_boundary_operator.py from the...	✓
`boundary_chi`	$\chi_\partial(x) = (x-1)(x^2-4x-1)$	characteristic polynomial of the companion matrix of the recurrence	
`boundary_growth`	$\nu = 2 + \sqrt{5} = \varphi^3$	Perron root of the boundary operator	
`boundary_field`	$\nu \in \mathbb{Q}(\sqrt{5})$, $\lambda^2 \in \mathbb{Q}(\sqrt{15})$	area and perimeter of the same tile inflate in different quadratic fields	
`boundary_dimension`	$\dim_{\text{box}} \partial = \log\nu / \log\lambda$	nu exceeds lambda, so the supertile boundary is fractal and the perimeter does not scale with the linear...	
`balanced_exponent`	$s^* = 1 - \log\nu / \log\lambda^2$	the thickness exponent making every substitution level carry comparable volume	
`compositum`	$\mu = \lambda^2/\nu$ has degree 4	the balanced ratio lies in Q(sqrt3, sqrt5) and in neither quadratic field alone, so the Spectre and the...	
`golden_absent`	$(x^2-4x-1) \nmid \chi_M$, χ_H	the boundary quadratic divides neither substitution matrix	
`lucas_deflation`	$\varphi^n :\ x^2 - L_n x + (-1)^n$	every power of the golden ratio has a Lucas quadratic for its minimal polynomial, so agreement of two such...	
`spectre_not_golden`	$\lambda^2 \neq \varphi^n$	the trace of the Spectre unit is eight, which is not a Lucas number, so the area inflation is not golden...	
`boundary_invariant`	$P(k+2) - 4P(k+1) - P(k) = -16$	the quadratic factor of chi applied to the sequence annihilates the growing part and leaves the constant...	✓
`boundary_growth_bound`	$P(k+3) \geq 4P(k+2) + 30$	the perimeter more than quadruples at every step past the seed, while lambda ¡ 4	✓
`lam_below_four`	$\lambda^2 < 16$	16 - lambdaˆ2 = 12 - sqrt15, positive since 12ˆ2 ¿ 15	✓
`boundary_exact`	$P(k)$ counted exactly in $\mathbb{Q}(\sqrt{3})$	the same sequence recounted with edges identified by equality of exact quadratic-field coordinates rather...	
`chair_mesh`	$(V, E, T) = (2138,\ 6408,\ 4272)$, $\chi = 2$	the 24 panels of the seven-cube chair, each cut by its 9x9 grid into 73 flat rectangles and 8 four-sided...	✓
`chair_volume`	$\text{vol} = 7 + \frac{4}{3} w^2 h \sum_s \gamma_s = 7$ for all w, h	the volume of the symbolic solid, for every base width w and height unit h at once: the signed heights sum...	✓
`chair_labels`	192 sites , $\gamma_s \in \{\pm 1, \ldots, \pm 12\}$, $\sum \gamma_s = 0$	height labels forced by exact complementarity on every shared face of the depth-two patch of the recorded...	✓
`chair_tilings`	$3^8 =$ 6561 rotation tilings of the doubled chair by 8 chairs	exact cover of the 56 cells of the doubled chair by eight rotated copies, by a bitmask search	
`chair_screen`	6555 tilings leave 4 free classes; 6 leave 12, and 3 of those use all 24 orientations	every one of the 6561 tilings is glued to depth two	
`chair_atlas`	2388 raw contacts → 44 legal → 30 realized	every face contact of two chairs (48 frames, translations in a box), the legal ones kept by exact site...	✓

Key	Claim	How	Lean
chair_equations	372 site equations , rank = 180 , dim ker = 12	the distinct complement equations c_A + c_B = 0 of the 30 realized contacts, ranked exactly over Q	✓
chair_chiral	Sym(decorated) = {1} , while $\lvert\mathrm{Sym(chair)}\rvert = 6 = 3 + 3$	the undecorated seven-cube chair has six symmetries, three of them mirrors, so it is achiral	
chair_forced_chirality	no mirror contact is legal: all 44 legal contacts have det = +1	of the 2388 raw face contacts 1194 place a mirror image against the root	
chair_orientation	$\chi(x) = x^3(x-8)(x-2)^{20}$ on the 24 rotation species	the orientation transfer matrix of the substitution on the 24 rotations	
chair_surface	surface $= 24 \cdot 4^N$, shared $= 12(8^N - 4^N)$, $B_c = 12 \cdot 4^N(2^c - 1)$	counted on the placed patch for N ¡= 4 with the face-directions exactly flat (4^(N+1) per +-x,+-y,+-z)	✓
chair_exponents	$D_b = 2$, $s^* = 1/3$, $\mu = 2$ ($\lambda = 2$, $\nu = 4$, $d = 3$)	the unified exponents D_b = log nu / log lambda, s* = 1 - D_b/d, mu = lambda^d / nu with the Chair44...	
chair_twist	supertile $= 2 \cdot$ chair , twist = 0 , $\frac{1}{8}\sum_i H_i = \frac{1}{4}I$	the 8 child chairs are laid on the doubled lattice and their cells compared with the doubled chair: equal,...	
chair_nets	surface = 24 squares , $(V, E, F) = (26, 48, 24)$, 261,434,325,600 spanning trees	the exposed unit faces of the seven-cube chair, their edge-adjacency graph (every edge shared by exactly...	
chair_slices	$\sum_j d_j = 2^n(1,1,1)$, layer interface $= 4^{n+1} - 4 \cdot 2^n$	the 8^n chairs of the depth-n patch, n¡=4, each reduced to the centre g(1,1,1)+t of its bounding cube and...	
orientation_sums	$S_n^{\mathrm{chair}} = 2^n$ exactly ; $\lvert S_n^{\mathrm{Spectre}}\rvert/\lambda^n \to 1.4489$	the sum of the orientation vectors over the depth-n patch: for the chair the z component of the notch...	
power_slopes	median $I(k)/N$ falls as $N^{-0.34}$ (Spectre) and $N^{-0.72}$ (chair)	the median over 300 fixed random wave vectors (seed 1) in the box [0, 2 pi / l]^d, with l the tile size...	
chair_coincidence	modular coincidence at depth 3 , $c(M) = 28 \cdot 2^M - 48$ $(M \geq 3)$	the cell substitution on 24 x 7 = 168 labelled unit cells, rebuilt from the child poses: column sum 8,...	✓
boundary_fibonacci	$P(k) = 2F_{3k+5} + 4$	the perimeter recurrence has seeds 14, 46, 182 and characteristic polynomial (x-1)(x^2-4x-1)	✓
boundary_directions	even $= F_{3k+4} + 1$, odd $= F_{3k+3} + 1$ $(k \leq 6)$	boundary edges of an order-k supertile, sorted by direction mod 180 degrees (the 14 edges point in...	✓
boundary_z3_bounded	max $\lvert\hat{h}(j)\rvert$ over the Z/3 and faithful modes $\leq \sqrt{13}$ $(k \leq 6)$	the DFT of the folded direction histogram over Z/6 = Z/2 x Z/3 (the split spectre_torsion uses for the six...	
centre_diffraction	$X_n = 2X_{n-1} \cup (\text{odd box} \setminus \text{corner octant})$; $\sum_k I(k) = \frac{1}{7} =$ density	the centre of every chair of the depth-n patch obeys the nested recursion above (checked against the...	
orientation_recursion	$S_{n+1} = C\,\overline{S_n}$, $\lvert S_n\rvert \sim c\,\lambda^n$, $c^2 = \frac{15+5\sqrt{3}+3\sqrt{5}+3\sqrt{15}}{20}$	the sum of exp(i angle) over the tiles of the depth-n supertile of each label obeys a linear recursion...	
orientation_channels	charpoly $= x^3(x-8)(x-2)^{20}$ (chair) ; $h_m = 2, 1, 0$ (Spectre)	the 24 orientation states of the Chair44 are split by the irreducible representations of the rotation...	
contact_matrices	Spectre: 14×14, rank 14, bipartite ; Chair44: 24×24, rank 24, not bipartite	the number of times native edge a of one tile meets native edge b of a neighbour, counted over the level-4...	
lift_integer	every vertex has integer coordinates $(p, q, r, s) \in \mathbb{Z}^4 = \mathbb{Z}[\zeta_{12}]$; rank = 4	every edge of the patch is a unit vector at a multiple of 30 degrees: even multiples are p+qw with w=exp(i...	
inflation_matrix	charpoly $= (x-1)^4(x+1)^4(x^4 - 8x^2 + 1)^2$, $\lambda^2 = 4 + \sqrt{15}$	the supertile recursion of spectre_patch moves four key points by rotations through multiples of 60...	
internal_plane	bounded plane over $\mathbb{Q}(\sqrt{15})$; $\dim_\partial$ of the window ≈ 1.2	a linear functional is bounded on the supertile corners exactly when it kills the expanding part...	